PyClima Explorer+

Intelligent Climate Data Platform

🎓 Student View

🔬 Researcher View

Team Algorythms | IIT (BHU) Varanasi · HACK IT OUT 2026

Turning raw NetCDF climate data into actionable intelligence.

- Map Atlas
- Time Studio
- ML Insights
- Compare Lab
- Story Mode
- 3D Orbit
- Dataset Intel

2,066	7	182	2 Lens
Time Steps (SST)	Feature Tabs	Events Detected	Student / Researcher

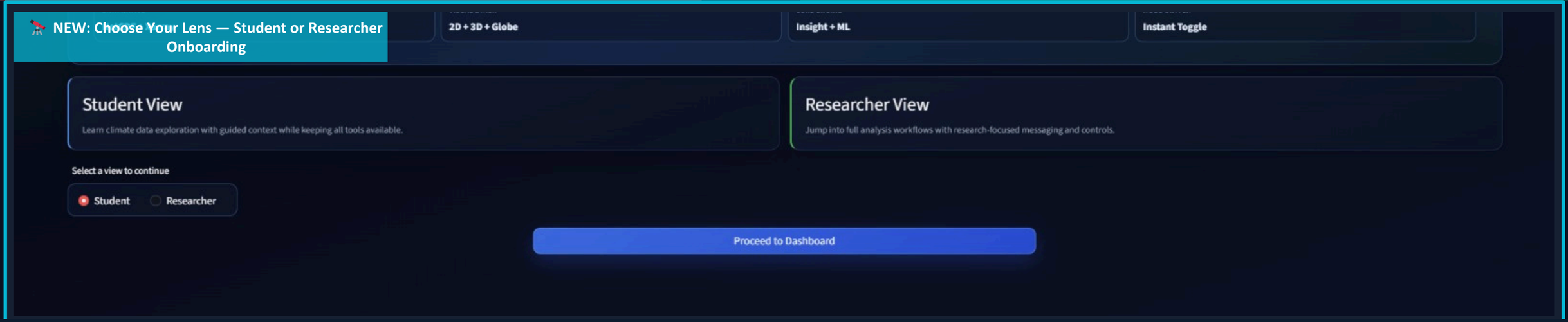
Problem Statement & Our Solution

❌ THE PROBLEM

- Climate NetCDF data is complex, multi-dimensional & inaccessible
- No single tool serves both researchers and general public
- Existing tools require command-line expertise or paid software
- Climate anomalies & record-breaking events go undetected

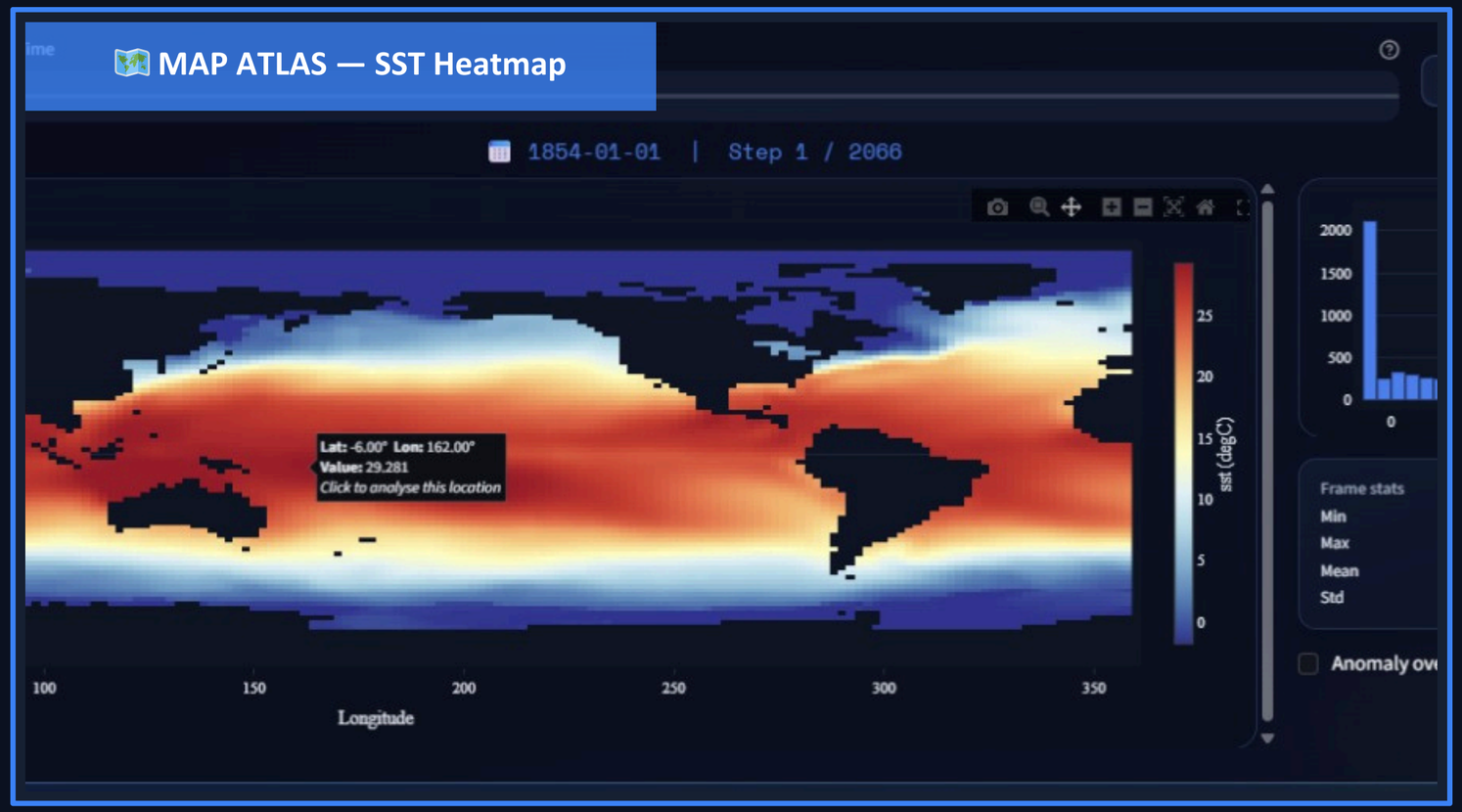
✅ OUR SOLUTION

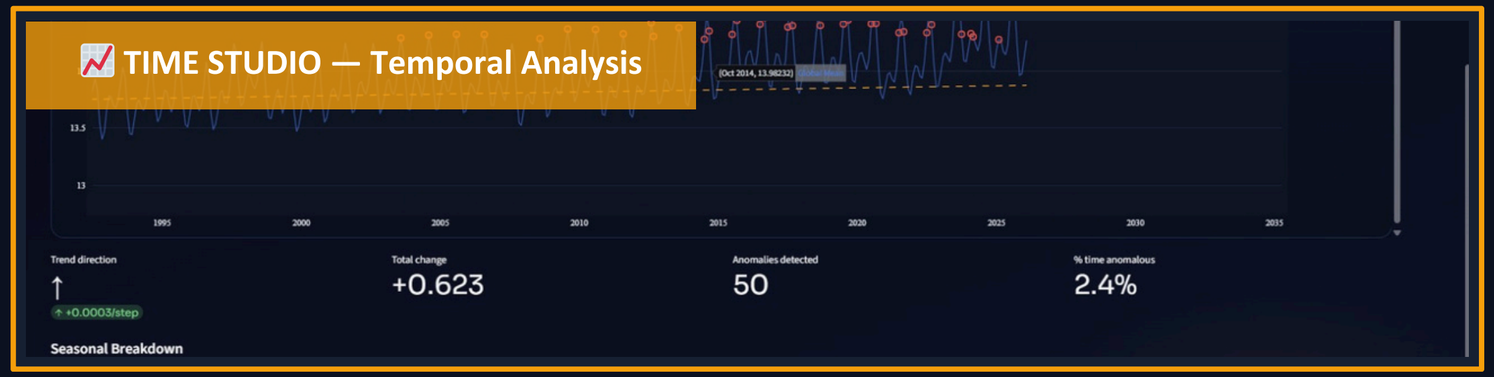
- PyClimaExplorer+ — drag-and-drop Streamlit dashboard for any .nc file
- Dual-lens onboarding: Student (guided) & Researcher (full power) modes
- 7 specialized tabs: Map Atlas, Time Studio, ML Insights, Compare Lab…
- ML Event Detector: 182 events, 33 record breaks, 71 rapid shifts auto-detected



Stack: Python · Streamlit · Xarray · Plotly · NumPy · Pandas · NetCDF

Map Atlas & Time Studio





Real NOAA SST Data
1854–2024 · 2,066 monthly time steps · global 89×180 grid

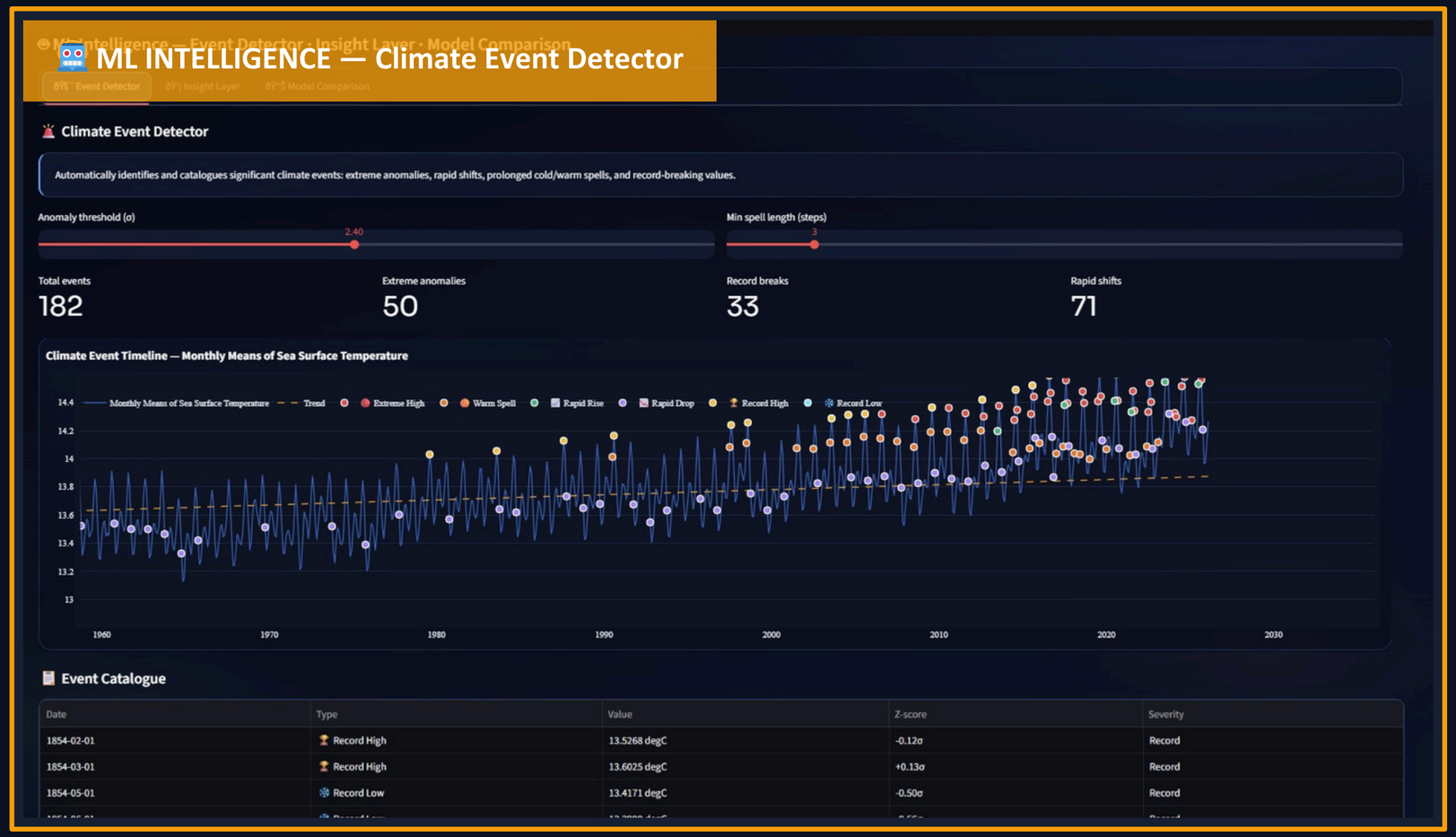
Click-to-Analyze
Click any point on map → instant local climate profile

Animated Time Slider
Slow / Normal / Fast animation with live frame counter

Live Frame Stats
Min / Max / Mean / Std updated every frame + histogram

↑ +0.623	50	2.4%	+0.0003
Total °C rise	Anomalies	Time anomalous	°C/step trend

ML Intelligence — Event Detector & Insight Layer



182
Total Events

50
Extreme Anomalies

33
Record Breaks

71
Rapid Shifts

Z-Score Anomaly Detection

Configurable threshold 1–4σ, marks extreme high/low events on timeline

Event Catalogue Table

Date, type, value, Z-score & severity for every detected climate event

Trend Projection

Linear trend fit with 30-step forecast & 95% confidence interval band

Bonus Features: 3D Orbit Globe & Compare Lab

Country Borders Overlay

Toggle coasts + country borders on the 3D globe

Point Cap Control

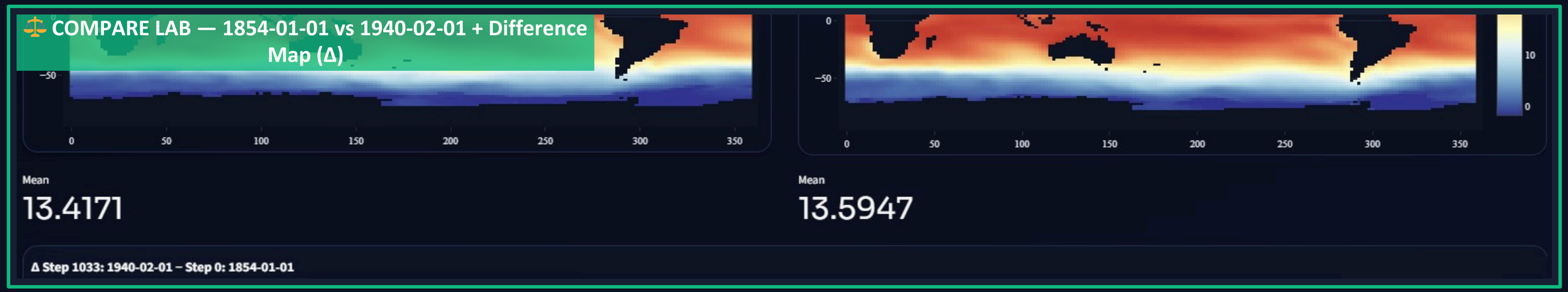
Slider 3,000–50,000 points for performance tuning

Animated Globe

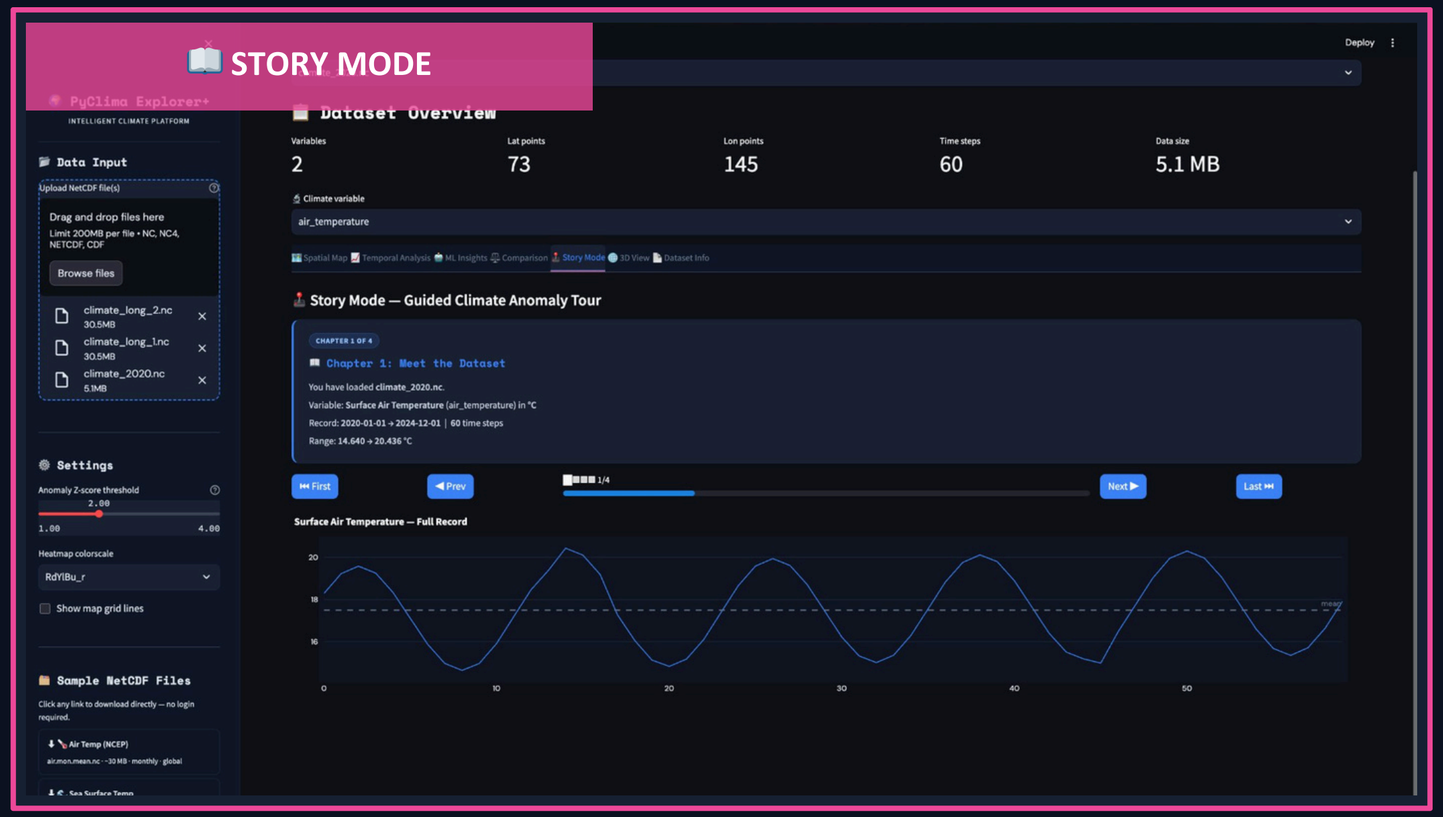
Play through all 2,066 time steps with animation controls

Multiple Projections

Orthographic, natural earth & more map projections



Story Mode & Dataset Intel



7-Chapter Guided Tour

Context → Dataset → Trend → Anomaly events → Outlook

Auto-Detected Events

Hottest/coldest anomalies auto-found and spotlighted

Spatial Anomaly Maps

Per-event maps show where extremes occurred globally

Student-Friendly Narrative

Context-aware text adapts to Student or Researcher lens

SST: Monthly Means Sea Surface Temp

Time: 2,066 steps 1854–01 → 2024

Grid: 89 × 180 -88° to 88° lat

Source: NOAA/NCEP ISST + HadISST

Thank You!

Live at :- https://vivek-patnaik25-pyclimaexplo-team-algorythms-app-pfxhjy.streamlit.app/

Team Algorythms

Vivek Patnaik · Ritesh Kumar Gouda · Anurag Panigrahi

What We Built

- All 3 core deliverables — Web UI, Spatial Map, Temporal Analysis
- All 3 bonus features — 3D Globe, Compare Lab, Story Mode
- ML Event Detector: 182 events, 50 anomalies, 33 records, 71 shifts
- Dual-lens onboarding: Student & Researcher modes
- Real NOAA SST dataset: 2,066 steps, 170 years of ocean data
- 7 specialised tabs — from Map Atlas to Dataset Intel